

Vertical Clarifier Floc Sweeping Equation Derivation

There is a zeroth order relationship between the number of successful collisions and the concentration of primary particles. Listed below is an extensive list of variable definitions. Following this variable list is a detailed derivation of the vertical clarifier floc sweeping equation.

Variable definitions:

$C_{coag_{particles}}$ = concentration of coagulant particles

C_p = concentration of primary particles

C_F = concentration of large flocs

N_c = number of successful collisions

N_F = number of large flocs

N_p = number of primary particles

α = attachment efficiency

v_r = relative velocity = $v_p - v_F$

A_p = projected area of large flocs in clarifier

A_T = total cross sectional area of clarifier

h = height of the floc blanket

R = radius of large flocs

r = radius of primary particles

$\theta_{flocculator}$ = hydraulic residence time of the flocculator

$\theta_{floc\ filter} = \frac{h}{v_p}$ = hydraulic residence time of the floc filter

ρ = density

(1)

$$\Delta N_c = \alpha \frac{N_p}{A_t h} v_r A_p \Delta t$$

Equation (1) describes the number of successful collisions over a unit of time as dependent on attachment efficiency, cleared volume by large flocs, and number of primary particles per floc filter volume. Attachment efficiency specifically refers to the probability that a large floc - primary particle collision is successful and sticks.

(2)

$$A_p = A_t \left(1 - \left(1 - \frac{\pi R^2}{A_t} \right)^{N_F} \right)$$

Equation (2) describes the total projected area of large flocs in the floc blanket based on total cross sectional area of the clarifier, projected area of one large floc (πR^2), and total number of large flocs in the floc blanket. The derivation for the total projected area of large flocs can be found [here](#). This equation accounts for vertical overlap between individual flocs by defining a summative series dependent on the total number of large flocs and probability of flocs overlapping.

(3)

$$N_F = \frac{C_F A_t h}{\rho_F \left(\frac{4}{3} \pi R^3 \right)}$$

Equation (3) defines the number of large flocs in the floc filter (N_F) as equal to the total mass of large flocs in the floc filter ($C_F A_t h$), divided by the mass of a single floc ($\rho_F \left(\frac{4}{3} \pi R^3 \right)$). The total mass of large flocs in the floc filter is defined by the concentration of large flocs multiplied by the volume of the floc blanket. The mass of a single floc is defined by the density of large flocs times the total volume of a single floc.

(4)

$$\frac{dN_c}{dt} = \frac{\alpha C_p}{\rho_p \left(\frac{4}{3}\pi r^3\right)} v_r A_T \left(1 - \left(1 - \frac{\pi R^2}{A_T}\right)^{\frac{C_f A_f h}{\rho_f \left(\frac{4}{3}\pi R^3\right)}}\right)$$

By substituting equation (3) and equation (2) into equation (1), $\frac{dN_c}{dt}$ can be defined in full as equation (4).

(5)

$$\frac{dC_p}{dN_c} = -k_1$$

In order to produce a differential equation relating the concentration of primary particles to time ($\frac{dC_p}{dt}$), equation (5) has been defined to relate the concentration of primary particles to the number of successful collisions as a 0th order relationship. See figure 1 for a graphical representation of equation (5). This relationship describes that every successful collision between a large floc and a primary particle reduces the concentration of primary particles by a constant amount.

Figure 1. Graph of Zeroth Order Relationship



(6)

$$\frac{dC_p}{dt} = \frac{-k_1 \alpha C_p}{\rho_p (\frac{4}{3} \pi r^3)} v_r A_T \left(1 - \left(1 - \frac{\pi R^2}{A_T}\right)^{\frac{C_F A_T h}{\rho_p (\frac{4}{3} \pi R^3)}}\right)$$

Combining equation (4) and (5) produces equation (6), a differential equation describing the change in concentration relative to the change in time for the mechanism of floc sweeping.

(7)

$$\frac{dC_p}{dt} = \frac{-k_1 \alpha C_p}{\rho_p (\frac{4}{3} \pi r^3)} v_r A_T$$

Equation (6) is not integrable if C_F is a function of C_p . It is known that for higher turbidity influent, the floc blanket will remain fully saturated. Therefore, the projected area, equation (2), can be simplified to $A_p = A_T$ when the influent turbidity is high, producing equation (7). The threshold for this assumption is not yet known, but will be calculated using future Plantita coagulant dosing data.

(8)

$$\int_{C_{p,flocculated}}^{C_{p,clarified}} \frac{1}{C_p} dC_p = \int_0^{\theta_{floc\ filter}} \frac{-k_1 \alpha C_p}{\rho_p (\frac{4}{3} \pi r^3)} v_r A_T dt$$

Taking equation (7) and integrating using separation of variables produces equation (8). In equation (8), the equation is being integrated where the change in C_p from the concentration after flocculation to the concentration after clarification corresponds to floc sweeping through the primary particles' residence time in the floc filter.

(9)

$$C_{p,clarified} = C_{p,flocculated} \exp\left(\frac{-k_1 \alpha v_r A_T \theta_{floc\ filter}}{\rho_p (\frac{4}{3} \pi r^3)}\right)$$

Equation (9) is the result of solving the integral defined in equation (8).

(10)

$$\bar{\alpha} = k_2 \frac{C_{coag_particle}}{C_{P,raw}}$$

In order to completely describe the change in concentration of primary particles due to floc sweeping, the attachment efficiency must be defined. As defined in equation (368) of the AguaClara textbook, the average attachment efficiency can be described in equation (10).

(11)

$$C_{P,clarified} = C_{P,flocculated} \exp\left(\frac{-k_1 k_2 v_r A_T \theta_{floc\ filter}}{\rho_p (\frac{4}{3} \pi r^3)} \frac{C_{coag_particle}}{C_{P,raw}}\right)$$

Replacing the attachment efficiency in equation (9) with equation (10) yields the final floc sweeping equation, equation (11). This equation defines that the concentration of primary particles after clarification is some fraction of the concentration of primary particles after flocculation, where the decrease in concentration corresponds to a combination of some unknown physical constants of the process, A_T the total area of the clarifier, $\theta_{floc\ filter}$ the hydraulic residence time of primary particles in the floc filter, the concentration of dosed coagulant particles, and the influent turbidity.

(12)

$$C_{coag_particle} = k_{pf} C_{P_0} (C_P^{-2/3} - C_{P_0}^{-2/3})$$

(13)

$$C_{coag_particle} = \frac{k_{pf}}{\theta_{flocculator}} C_{P,raw} (C_{P,clarified}^{-2/3} \exp\left(-\frac{2}{3} \frac{k_1 k_2 v_r A_T \theta_{floc\ filter}}{\rho_p (\frac{4}{3} \pi r^3)} \frac{C_{coag_particle}}{C_{P,raw}}\right) - C_{P,raw}^{-2/3})$$

Substituting the $C_{P,flocculated}$ of equation (11) into equation (12) produces the cumulative equation (13) that describes the change in turbidity of the water due to flocculation, clarification,

and floc sweeping as a function of raw turbidity and coagulant dosing. Equation 12 comes from the AguaClara textbook [\(372\)](#).

(14)

$$k_{fs} = \frac{-2k_1 k_2 v_r}{3\rho_p(\frac{4}{3}\pi r^3)}$$

All of the unknown physical constants (excluding known constants such as A_T) can be treated as a cumulative constant k_{fs} in equation (14).

(15)

$$C_{coag_particle} = \frac{k_{pf}}{\theta_{floculator}} C_{P,raw} (C_{P,clarified}^{-2/3} \exp(k_{fs} A_T \theta_{floc\ filter} \frac{C_{coag_particle}}{C_{P,raw}}) - C_{P,raw}^{-2/3})$$

As a result, equation (13) simplifies into the final equation describing flocculation, clarification, and floc sweeping, equation (15).